

CS402 · COMMITTEE PRESENTATION · 12 MIN

Earthquake Damage Assessment & Assistance Needs via Social Media

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● THE PROBLEM

The infrastructure you need is the infrastructure that fell.

- **Kahramanmaraş, Feb 2023** — cell towers down, roads impassable, official reporting stalled for hours.
Field teams still had to mobilise and survey by hand.
- In that window, coordinators allocate scarce resources with **no verified ground truth**.
- Citizens post anyway — **before** any formal assessment is possible.

● WHY THIS GAP, SPECIFICALLY

Social media as first responder — but not in Turkish, and not live.

- Twitter damage-signal work is established: Ridgecrest, Australian floods, disaster imagery — posts correlate with severity and surface local needs.
- Almost all of it is **single-platform, English-language, post-hoc**.
- Turkish is agglutinative — sparse token space breaks standard NER/classification assumptions. Prior Turkish NER work is morphology-aware but never applied to **live disaster response**.

● DATA SOURCE

Ekşi Sözlük: a forum with editorial gates

- Since 1999. Authorship requires **approved sample entries** — a real barrier most social platforms don't have.
- Enforced Turkish orthography, but a mandatory **all-lowercase** convention that breaks capitalisation-based NER heuristics.
- Twitter/X API pricing made live monitoring at scale a non-starter for this project.
- 6 Feb 2023 main thread: **760** entries in hour one, **4,260** in 24h. Relief-coordination thread: **10,000+** in a week.

"more grammatically regular than the latter, yet far from a solved NLP problem"

● CORPUS & LABEL TAXONOMY

Five earthquakes, two decades, one taxonomy built bottom-up

18,628

entries scraped, 5 events

8,222

labelled entries

~90%

from Kahramanmaraş 2023

34.5

mean tokens / entry

H Damage

Y Help request

B General info

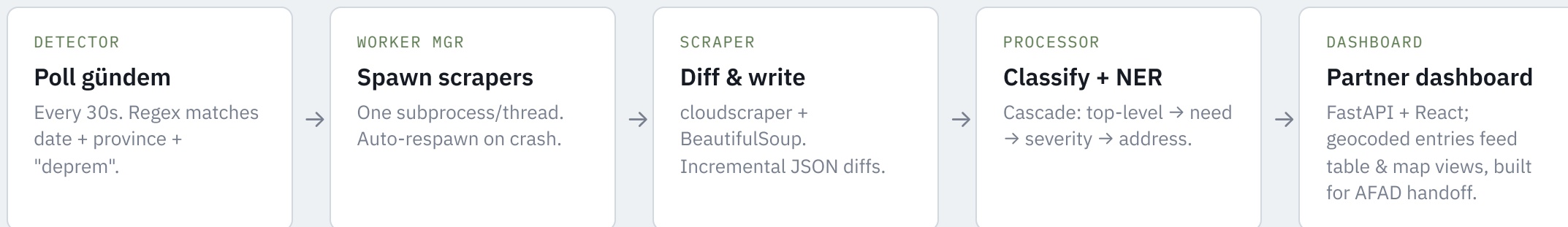
N Not relevant

ÇH/OH/AH severity

11× need-type labels

● SYSTEM ARCHITECTURE

Four daemon threads, one live pipeline



● METHODOLOGY

A cascade of fine-tuned Turkish BERTs

- **Top-level (H/Y/B):** three independent binary classifiers, not one softmax — lets an entry carry compound labels like HY.
- **Need-type:** one multi-label head, 11 logits, BCE loss — rescue and food/water can both fire.
- **Severity:** also fine-tuned, same recipe — benchmarked against keyword and BM25 baselines just like need-type, and it wins where they collapse: OH (moderate) has just 6 test samples.
- **Address NER:** token classification over 8 span types, segmented into multiple addresses per entry, resolved via Google geocoding.

● JUSTIFYING THE COST OF FINE-TUNING

Was BERT worth it over something cheaper?

- Benchmarked need-type *and* severity classification against **keyword matching** and **BM25 retrieval**; need-type also against **zero-shot Claude Haiku** — same protocol, same test split.
- Framed as a direct question to the committee: is a fine-tuned transformer defensible for an 11-category task at this data volume?

● RESULTS — TOP-LEVEL & SEVERITY

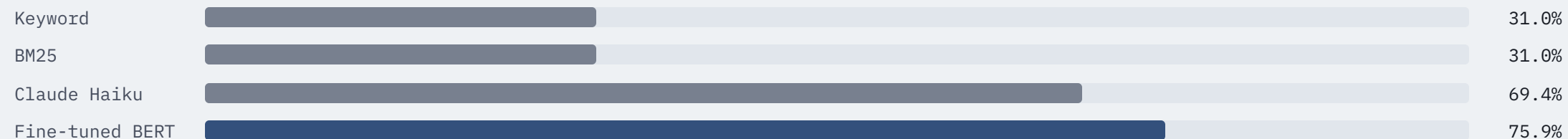
96.0% accuracy, 91.9% macro-F1

CLASSIFIER	ACC.	PREC.	REC.	F1
H — damage	97.5%	94.3%	92.7%	93.5%
Y — help req.	95.1%	85.8%	85.2%	85.5%
B — gen. info	95.2%	82.6%	77.2%	79.8%



● RESULTS — NEED-TYPE & ADDRESS

Fine-tuning beats a zero-shot LLM by 6.5 F1 points



micro-F1, 11-category multi-label

- Address NER: **79.5%** precision, **88.7%** recall, **83.3%** span-F1.
- Quality tracks the data: 2023 authors gave street + building + floor; earlier events gave city/district only.

● DISCUSSION

The hard part isn't the language. It's the label boundary.

- A zero-shot LLM already recovers most of the need-type signal — so the text isn't unreadable.
- BERT's edge is learning **overlapping category boundaries** that even human annotators contest.
- Address extraction is bounded by **annotation coverage**, not model capacity — it degrades exactly where street-level detail is absent.

● CONCLUSION & FUTURE WORK

Operational today. Two clear next steps.

- Detects seismic events within seconds, classifies and address-tags continuously, surfaced on a live dashboard built for AFAD handoff.
Access controls and KVKK-compliant data-sharing land before that handoff.
- **Turkish disaster WordNet** — extends KeNet with domain vocabulary, benefits both classification and NER.
- **Broaden address annotation** beyond 2023-style entries to smaller, sparser events.

END OF PRESENTATION · 11:30

Questions?

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